Model 1: Penalized Linear Regression (LASSO, Ridge, Elastic Net)

Purpose

Model 1 asks whether behavioral characteristics predict 1-year percent weight change beyond demographics and starting BMI, and if so, which behaviors matter most. The sample is 2,794 U.S. adults who tried to lose weight (NHANES Aug 2021–Aug 2023). The outcome is pct_wt_change, the percent change from weight one year ago to current weight. Negative values mean weight was lost.

Predictor sets

Each model is fit twice:

- **Covariates only (15):** age, sex, race/ethnicity, education, income-to-poverty ratio, marital status, household size, U.S.-born, veteran status, and BMI one year ago.
- **Covariates + behaviors (27):** the 15 covariates plus 12 behaviors: moderate and vigorous activity (min/week), sedentary hours/day, depressive symptoms (PHQ-9), drinks/week, sleep hours, smoking status, and meal-planning and food-shopping roles.

Comparing the two versions shows how much predictive value the behaviors add.

Steps

1. **Train/test split.** The data are split 80/20 (2,235 training, 559 test). The split is balanced on the outcome so both sets get a similar range of weight change. The test set is used only for the final evaluation.
2. **Preprocessing pipeline.** This runs inside the model fitting, so nothing from the test set leaks into training.
   - Missing values are filled with the median from the training data.
   - Income, PHQ-9 and drinks/week are each missing for about 12% of people. These three also get a yes/no "missing" flag, so the model can learn whether not answering is itself informative.
   - All predictors are standardized to mean 0 and SD 1, which lets coefficients be compared directly.
3. **Three penalized regressions.** The penalty strength (and the L1/L2 mix for elastic net) is chosen by 10-fold cross-validation on the training set.
   - **LASSO (L1 penalty)** shrinks coefficients and sets weak predictors to exactly zero, so it also does variable selection.
   - **Ridge (L2 penalty)** shrinks all coefficients toward zero but keeps every predictor. It handles correlated predictors well.
   - **Elastic net** mixes the two penalties.
4. **Performance.** Models are scored on the test set with R^2, RMSE and MAE, and compared against a baseline that predicts the average for everyone.
5. **Do behaviors add value?** This is checked in two ways:

  - A paired bootstrap on the test set (2,000 resamples) gives the change in R^2 and RMSE with 95% confidence intervals.
  - Repeated 5-fold cross-validation (5 repeats, 25 folds) on the training set gives a more stable estimate that depends less on one split.
6. **Which predictors matter?**
  - **Standardized coefficients** give the change in % weight per 1 SD of each predictor.
  - **The LASSO path** shows the order in which predictors enter the model as the penalty is relaxed. Predictors that enter earlier carry stronger signal.
  - **Bootstrap selection stability:** LASSO is refit on 200 bootstrap samples, recording how often each predictor is selected and whether its sign stays the same. Predictors selected in most resamples with a consistent sign are the most reliable findings.
7. **Residual check.** Plots of observed vs. predicted values and of residuals check the model's fit.

Key results

Model	Predictors	Test R^2	Test RMSE
LASSO	Covariates	0.074	8.74
LASSO	+ Behaviors	0.080	8.71
Ridge	Covariates	0.072	8.75
Ridge	+ Behaviors	0.079	8.72
Elastic net	Covariates	0.075	8.74
Elastic net	+ Behaviors	0.080	8.71

- **Behaviors add a small, fairly consistent gain.** In cross-validation, R^2 went up by about 0.008 with behaviors, which improved the model in 20–21 of 25 folds. On the single test set the gain was 0.005–0.008, and the 95% CIs include zero (e.g., LASSO $\Delta R^2 = 0.005$, CI −0.010 to 0.021). So the improvement is not statistically certain.
- **Starting BMI dominates.** It has by far the largest coefficient (about −3.0% per SD): people with higher starting BMI lost more weight in percentage terms.
- **The most robust behavioral predictors** were each selected in at least 88% of bootstrap resamples with a consistent sign:
  - **More weight loss:** being the household's main meal planner, current smoking, more vigorous activity.
  - **Less weight loss:** more sedentary hours, higher depressive symptoms (PHQ-9).
  - Missing drinks/week data was also linked to less weight loss. This may reflect how the alcohol questions were skipped, so it shouldn't be read as a behavioral effect.
- **Overall explanatory power is modest** (R^2 about 0.08–0.12). That's expected for self-reported 1-year weight change, which is noisy.

Limitations

- The coefficients are **associations, not causal effects**. The data are cross-sectional.

- Behaviors were measured at the exam, at the **end** of the weight-change period, so the direction of influence is unclear. For example, depression could be a result of weight change rather than a cause.
- The models are **unweighted**, so the results describe this sample rather than the U.S. population. NHANES survey weights can be applied by setting USE_SURVEY_WEIGHTS = True in the notebook.